Introduction to Machine Learning

CISC 484/684

February 20, 2026

1 Project Overview

The semester project provides hands-on experience with machine learning through methodological analysis of domain-specific data or reproduction of state-of-the-art methods. You will work in teams of **3-5 students** to design, implement, and evaluate a complete machine learning pipeline.

2 Project Requirements

2.1 Team Composition & Effort

- Teams may consist of 3-5 students (no solo projects)
- All team members must contribute substantially to all phases
- Cross-disciplinary teams encouraged for diverse perspectives

2.2 Data Requirements

To ensure meaningful analysis, your dataset must meet **one** of the following criteria:

- **High-volume tabular:** $\geq 10,000$ instances with ≥ 20 features
- **High-dimensional:** ≥ 200 instances with $\geq 10,000$ features
- **Structured data:** For images, time series, or text: sufficient instances for training/validation/test splits (minimum 1,000 total)

2.3 Scientific Rigor Requirements

- **Hypothesis-driven:** Clearly state your research question/hypothesis
- **Performance metrics:** Define appropriate evaluation metrics
- **Experimental design:** Design controlled experiments to test your hypothesis
- **Reproducibility:** Document all steps for reproducibility

3 Project Types & Examples

Select **one** primary project type (examples not exhaustive):

Type	Description & Research Questions
Scaling Analysis	Test how performance and computational requirements vary with sample size, features, classes, or model complexity. Compare algorithms of varying complexity across multiple datasets.
Applied Analysis	Apply existing methods to novel data. Which approach performs best under what conditions? Do models fail on the same examples?
Method Extension	Combine non-trivial techniques (preprocessing, regularization, optimization, uncertainty quantification) with existing methods. What problem does your extension solve?
Reproducible Research	Reproduce state-of-the-art methods on original datasets, then apply to novel data. Is there evidence for consistent method ranking?
Novel Benchmarking	Perform comprehensive survey of methods for a specific task. Establish new benchmarks with clear performance frontiers.

4 Methodological Approaches

Projects should incorporate techniques from at least two categories:

4.1 Exploratory Methods

Clustering, dimensionality reduction, topological data analysis, feature selection, explainable AI (XAI), uncertainty quantification.

4.2 Learning Paradigms

Supervised/unsupervised learning, semi-supervised/active learning, representation learning (self-supervised, contrastive, VAEs), generative modeling (GANs, diffusion models, flows), time series analysis, reinforcement learning, domain adaptation, multi-task learning.

4.3 Technical Components

Meta-optimization (hyperparameter tuning, NAS), kernel methods, graph-based methods, ensemble methods, matrix/tensor methods, missing data techniques.

5 Data Sources

- **General Repositories:** UCI ML Repository, OpenML, Kaggle
- **Domain-Specific:** PhysioNet (clinical), remote sensing datasets, financial analytics, neuroimaging (Neurosynth), -omics data
- **Other:** Audio/signal data, natural images, game environments, geolocation data, document collections

6 Project Timeline & Deliverables (100 points total)

Phase	Deliverable & Description	Points
Proposal	**Project Proposal:** Project definition, data description, EDA plan, methodology outline. **Can be revised once** based on feedback.	10
Execution	**Preliminary Results:** Baseline models, initial analysis, progress update	20
	Weekly Updates: Regular progress reports & team meetings	10
Final	**Final Report:** Complete analysis, error analysis, conclusions (8-12 pages)	45
	Final Presentation: 15-minute presentation + Q&A	15
Total Project Points		**100**

Note: Project points constitute 20% of your final course grade.

7 Technical Roadmap

7.1 Phase 1: Exploratory Data Analysis & Baselines

- Conduct thorough EDA to understand data characteristics and challenges
- Establish theoretical floor with classical algorithms (linear models, decision trees)
- Implement competitive baselines (XGBoost/LightGBM for tabular, SVMs/kernel methods for structured data)
- Discuss inductive biases and appropriateness for your problem

7.2 Phase 2: Advanced Method Implementation

- **Explore advanced architectures** appropriate for your data type. This could include:
  - **Computer Vision:** CNNs, ResNet, Vision Transformers, EfficientNet
  - **NLP/Sequences:** Transformers, BERT, LSTMs, GRUs, RoBERTa, GPT variants
  - **Generative Models:** GANs, VAEs, Diffusion Models, Normalizing Flows
  - **Graph Data:** GCNs, GATs, GraphSAGE, Graph Transformers
  - **Time Series:** Temporal Convolutional Networks, TCNs, N-BEATS

  These are examples only - you should explore architectures relevant to your specific problem domain.
- Justify architectural choices with respect to baseline performance
- Include proper validation strategies (cross-validation, hold-out sets)
- Compare against established baselines to quantify performance improvements

7.3 Phase 3: Comprehensive Analysis

- **Error Analysis:** Identify failure modes and patterns
- **Uncertainty Quantification:** Assess model confidence and reliability
- **Ablation Studies:** Test importance of model components
- **Statistical Testing:** Validate performance differences

8 Final Presentation Details

- **Undergraduate/Graduate Students:** Presentation mandatory (included in 15 points)
- **Format:** 15-minute presentation + 5-minute Q&A
- **Evaluation:** Technical sophistication, clarity, quality of execution, response to questions

Grading Rubric (Final Report)

- **Scientific Rigor (15 pts):** Clear hypothesis, experimental design, statistical validity.
- **Technical Execution (15 pts):** Code quality, methodology implementation, reproducibility.
- **Analysis Depth (10 pts):** Error analysis, uncertainty quantification, insightful conclusions.
- **Clarity & Presentation (5 pts):** Organization, writing quality, figure/tables.